CERTIFICATE

Certified that this project report **"A Machine Learning-Based Framework For Real-Time Detection Of Prompt Injection Attacks In Llm-Integrated Applications"** is the bonafide work of **Yash Chavan** and **Sohel Sheikh**, who carried out the project work under the supervision of Dr. Abhishek D. Pathak in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Cyber Security) during the Academic Year 2026-27.

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ABSTRACT

CHAPTER 1: INTRODUCTION

Large Language Models (LLMs) are increasingly embedded in customer-facing chatbots, coding assistants and autonomous agents, and these applications pass untrusted user text directly to the model. The resulting vulnerability, prompt injection, rewrites the behaviour of the model by natural-language instruction alone. This chapter introduces the problem, surveys the research literature between 2024 and 2026, defines the scope of the work and the hardware and software environment, and identifies the evaluation gap: most published detectors are validated only on same-dataset splits rather than on genuinely unseen attack distributions.

CHAPTER 2: PROPOSED SYSTEM

The proposed system, PromptShield, is a model-agnostic firewall that screens every prompt before it reaches any OpenAI-compatible LLM. It combines a TF-IDF linear classifier (50,039 input features, of which 39 are hand-engineered heuristics) trained on 23,357 public prompt injection samples, with thresholded verdicts: ALLOW, FLAG (72%) and BLOCK (74%). The chapter states the objectives of the proposed system, specifies the user requirements, presents the system architecture, and reports the evaluation metrics, including a test verdict accuracy of 98.26% and an AUROC of 0.9987.

CHAPTER 3: ANALYSIS AND DESIGN

The analysis centres on two relational entities, users and events, and eight cooperating modules: authentication, the detection engine, the output firewall, the protected chat application, attack logging, the security dashboard, automated PDF reporting and a browser extension that scans prompts inside third-party chat sites and stops blocked prompts from being sent. Entity relationship and data flow diagrams and table designs are presented, followed by annotated screenshots of the working deployment, which caught 30 of 30 curated red-team attacks and 29 of 29 live stress-test attacks.

CHAPTER 4: CONCLUSION

The project demonstrates that a lightweight, explainable TF-IDF-plus-heuristics ensemble provides production-quality prompt injection defence when paired with an output firewall and per-account audit trail.

Conclusions are drawn on effectiveness, misuse resistance and deployment cost; future work covers adaptive attacks, semantic encoders, multilingual coverage and feedback-assisted retraining; and the limitations of a lexically driven detector are stated honestly.

Keywords: prompt injection, large language models, TF-IDF, heuristic ensemble, output firewall, AUROC, browser extension, FastAPI, cyber security.

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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

API - Application Programming Interface

AUROC - Area Under the Receiver Operating Characteristic curve

B.Tech - Bachelor of Technology

CDP - Chrome DevTools Protocol

CSV - Comma Separated Values

DB - Database

DFD - Data Flow Diagram

ER - Entity Relationship

FK - Foreign Key

HMAC - Hash-based Message Authentication Code

HTTP - HyperText Transfer Protocol

JSON - JavaScript Object Notation

LLM - Large Language Model

NAACL - North American Chapter of the Association for Computational Linguistics

NLP - Natural Language Processing

PBKDF2 - Password-Based Key Derivation Function 2

PDF - Portable Document Format

PK - Primary Key

RDBMS - Relational Database Management System

ROC - Receiver Operating Characteristic

SDK - Software Development Kit

TF-IDF - Term Frequency - Inverse Document Frequency

UI - User Interface

USENIX - UNIX Users Group Association Symposium Series

UUID - Universally Unique Identifier

WAF - Web Application Firewall